

Joseph-Vu Trinh

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Education

Virginia Tech

Bachelor of Science - Computer Science

Blacksburg, VA

Graduating May 2028

Experience

Software Developer

June 2026 – Present

Virginia Tech Department of Computer Science

Blacksburg, VA

- Shipped backend REST API services in **Python** and **Flask** for DBWorkout, a SQL practice platform deployed to **170** students in Virginia Tech's CS4604 database course, where **90%** reported improved learning value post-deployment.
- Executed student queries against isolated **PostgreSQL** sandboxes via SQLAlchemy, diffing result sets against expected output to grade submissions automatically instead of by hand
- Added **OpenAI API** hint generation from a student's failed query and exercise schema, driving a feedback overhaul after only **42%** of surveyed students rated the original feedback actionable.

Full Stack Developer Intern

Sep 2024 – Dec 2024

Engauge

College Park, MD

- Built a collaborative whiteboard in **React** and **TypeScript** with Firestore persistence, synchronizing state across **30+** concurrent clients
- Modeled Firestore documents to scope whiteboards to individual students while giving instructors visibility across session submissions
- Rebuilt the front-end component architecture of a classroom web application, cutting re-renders for sessions of **150+** students
- Implemented a live transcription and chat feature, wiring WebSocket updates and **GCP Cloud Functions** to persist transcripts and conversation history

Undergraduate Researcher

Jan 2026 – June 2026

Virginia Tech Department of Computer Science

Blacksburg, VA

- Engineered a machine learning pipeline in **Python** using **OpenPose** to extract skeletal pose and gaze features from **180** one-minute classroom video segments across **300+** students.
- Classified student engagement from the extracted features without storing identifiable imagery, improving classification accuracy by **25%** over the prior approach.
- Benchmarked LLM-generated engagement observations against human annotations using the **OpenAI API**, reducing malformed model responses by **85%**.

Web Developer

July 2026 – Present

SASE at Virginia Tech

Blacksburg, VA

- Rebuilding the club site in **Next.js**, **TypeScript**, and **Tailwind CSS**, replacing a legacy site with a mobile-first architecture
- Working with organization leadership to turn requirements into designs, maintaining the live site with event listings and image updates each semester

Projects

Professor Search Platform | VTHacks | Python, React, Databricks Agent Bricks, GraphQL

- Scraped and ingested **16,000** records spanning **577** professors and **3,267** reviews through asynchronous Reddit collection and paginated **GraphQL**, with filtering, deduplication, retry logic, and rate-limit handling.
- Integrated the curated dataset with **Databricks Agent Bricks** for an LLM-powered recommendation system answering natural-language questions about professors and courses, and developed the supporting React interface

Playlist Curator | React, TypeScript, FastAPI, Anthropic API, Spotify Web API, librosa

- Developed a playlist curation tool in **React**, **TypeScript**, and **FastAPI** that scores Spotify tracks against user-defined themes for review before applying changes.
- Analyzed playlists of arbitrary size through **10+** backend endpoints and local audio signal analysis with librosa, combining acoustic features with Spotify metadata.

Technical Skills

Languages: Python, Java, C/C++, C#, SQL, JavaScript, TypeScript, Swift

Frameworks: Flask, FastAPI, React, Node.js, Tailwind CSS, React Native, SwiftUI

Tools & Technologies: Git, Linux/Unix, Figma, PostgreSQL, Docker, CI/CD, REST APIs, AWS, Firebase/Firestore, MongoDB (NoSQL), JUnit